

BAAZ JKBOSE



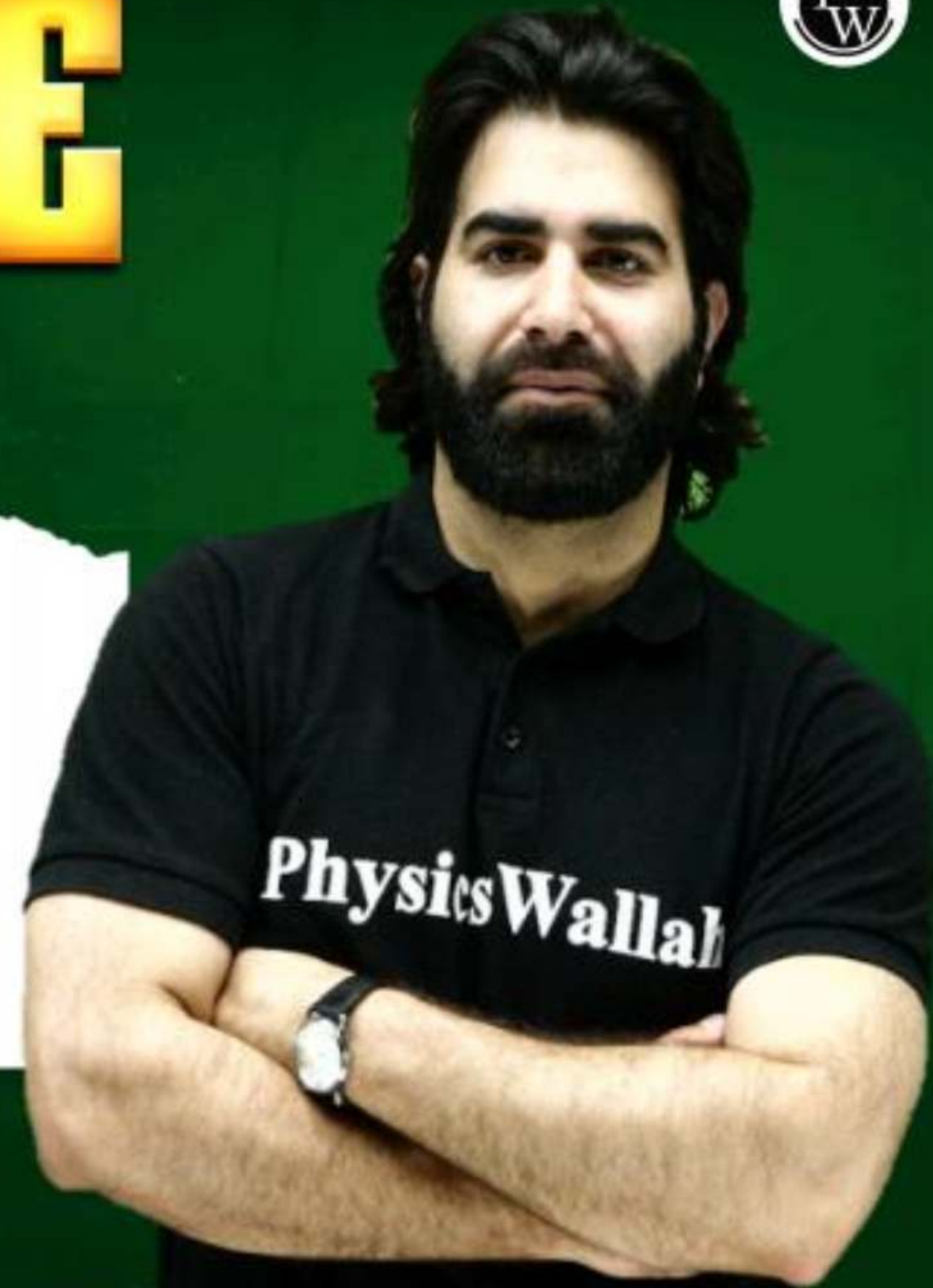
2026 CLASS-11TH

Breathing and Exchange of Gases

LECTURE-1

Zoology

Dr Adil Altaf sir



Topics To Be Covered

- 1 Introduction
- 2 Types of respiratory organs
- 3 Human Respiratory System

Physiology

↳ Gross Functioning

→ Systems - Functions

- Respiratory System
- Circulatory System
- Excretory System
- Musculoskeletal System
- Nervous System
- Hormonal System

Bulk Syllabus

Doc Fox

Task (Sincerely)

- Listen
- Understand
- Note Making
- Revision
- DPP
- Test

Breathing and (Gaseous Exchange)

Breathing

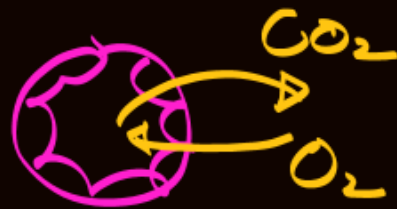
↳ Act involving
(Process)

involving inspiration and Expiration.
Breath in Breath out

eg

Paramecia

↳ Does not
have lungs



'Definition of Breathing'

→ Can be explained also - It is a process in which organism exchanges gases between Atmosphere and themselves.

- NCERT - Breathing is commonly known as Respiration (Physical Respiration).



CHAPTER 14

BREATHING AND EXCHANGE OF GASES

As you have read earlier, oxygen (O_2) is utilised by the organisms to indirectly break down simple molecules like glucose, amino acids, fatty acids, etc., to derive energy to perform various activities. Carbon dioxide (CO_2) which is harmful is also released during the above catabolic reactions. It is, therefore, evident that O_2 has to be continuously provided to the cells and CO_2 produced by the cells have to be released out. This process of exchange of O_2 from the atmosphere with CO_2 produced by the cells is called **breathing**, commonly known as **respiration**. Place your hands on your chest; you can feel the chest moving up and down. You know that it is due to breathing. How do we breathe? The respiratory organs and the mechanism of breathing are described in the following sections of this chapter.

Difference between Breathing & Respiration.

— Breathing (ventilation)

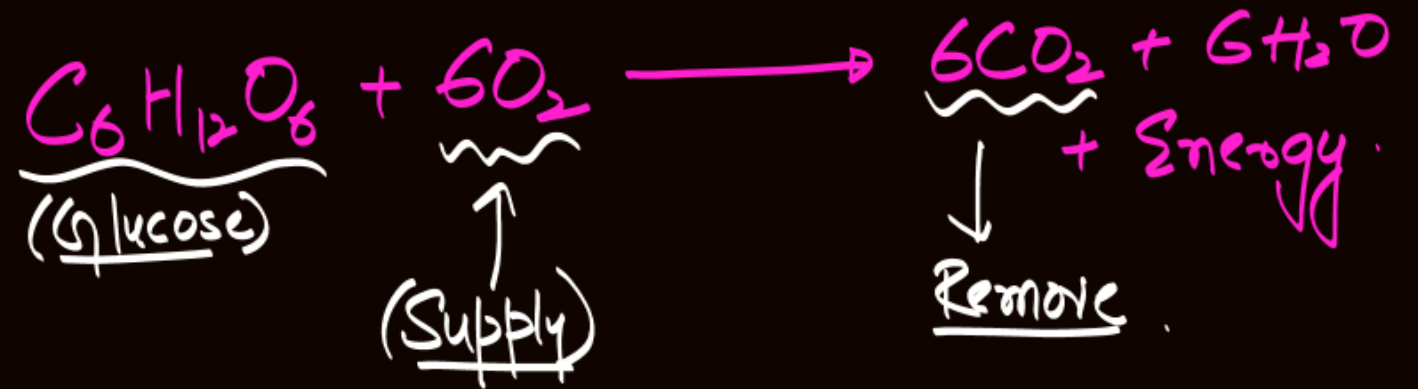
→ Exchange of gases

→ (Mechanical Process)

In which gases are removed and supplied.

Respiration

→ Oxidative Breakdown of food to give Energy.



→ Supplying oxygen and Removing CO_2 , we need Breathing.

→ That means, Breathing is a process, which comes under Respiration.

→ Respiration is of two parts

① Physical Respiration (Exchange of gases).

② Chemical Respiration (Actual Breakdown of food).

Respiration → can be Anaerobic and can be Aerobic.
(O₂ Not Required) (O₂ is Required)

→ it includes - Glycolysis, Krieb's Cycle etc

Detail
Plant Physiology
(Respiration)

→ Types of Respiratory organs in Different organism.

- <u>Porifera, Coelenterata, Flatworms</u>	→ Respiration through <u>Body Surface</u> . 
 (Tapeworm)	
- <u>Annelida (Earthworm)</u>	→ Through <u>Moist Skin</u>  H_2O 
- <u>Arthropoda</u> → <u>Aquatic (Prawn)</u>	→ <u>Gills (Branchial)</u> ↳ (<u>Cutaneous Respiration</u>)
→ <u>Terrestrial (Cockroach)</u>	→ <u>Tracheal System</u>  → Tracheal system.
→ <u>Mollusca</u>	→ <u>Gills</u> .
→ <u>Fishes</u>	→ <u>Gills</u> .
→ <u>Amphibian</u>	→ (Skin, Buccopharyngeal, <u>Gills</u> (Tadpole), lungs).
→ <u>Mammals, Aves, Reptiles</u> (Human, Cat, Dog, Cow, Rat).	- <u>Lungs (Pulmonary Respiration)</u> .

Qno 1) Respiratory organ in earthworm is :

- (a) Lungs
- (b) Tracheal System
- (c) Moist skin (✓)
- (d) Gills

Questions → New Pathway

→ Buccopharyngeal.

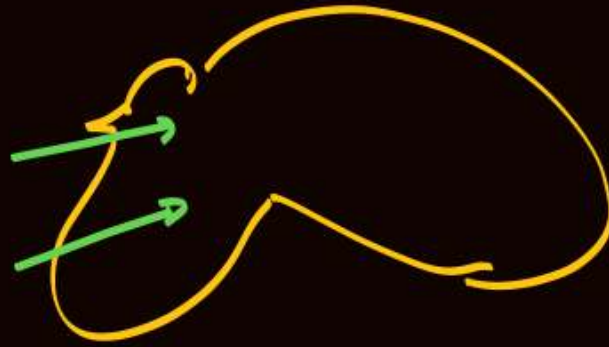
↓
Buccal
Cavity
(oral C)

↓
pharynx

→ (Prog)

→ (Outcome)

- Gills → Branchial Respiration.
- Skin → Cutaneous "
- Lungs → Pulmonary "



Human Respiratory System

↳ It is that system which is used in Breathing (Physical Respiration)

- It includes;

① Nose (✓)

② pharynx.

③ Larynx.

④ Trachea.

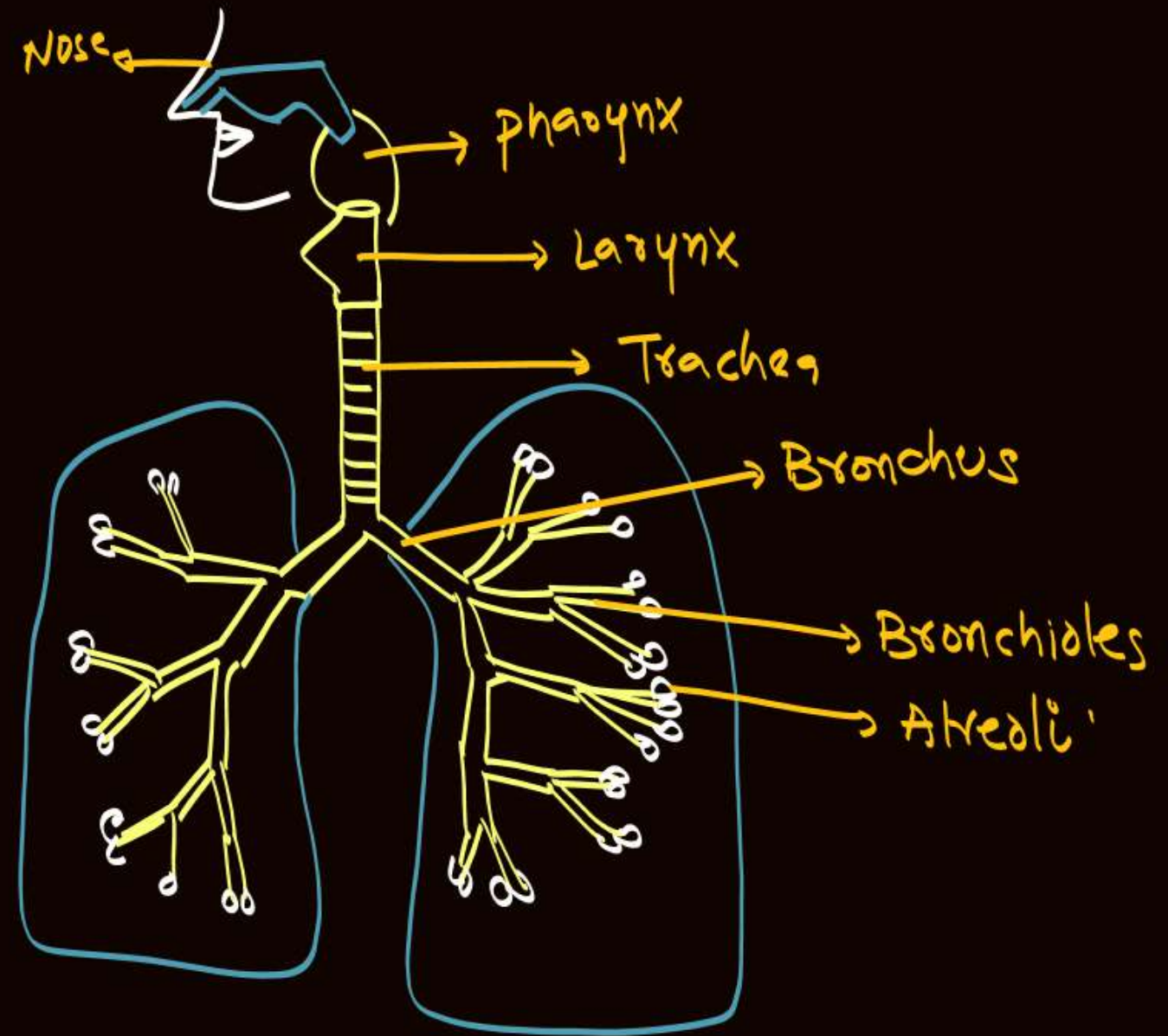
⑤ Bronchi.

⑥ Bronchioles.

⑦ Alveoli.

Lungs

Rough Diagram.



Nose

↳ It is an organ which is partly Bony and partly cartilagenous.

→ Bony Part - Two Nasal Bones

→ Cartilagenous Part → Two types (wall).

(a) Nasal Septum (Hyaline)

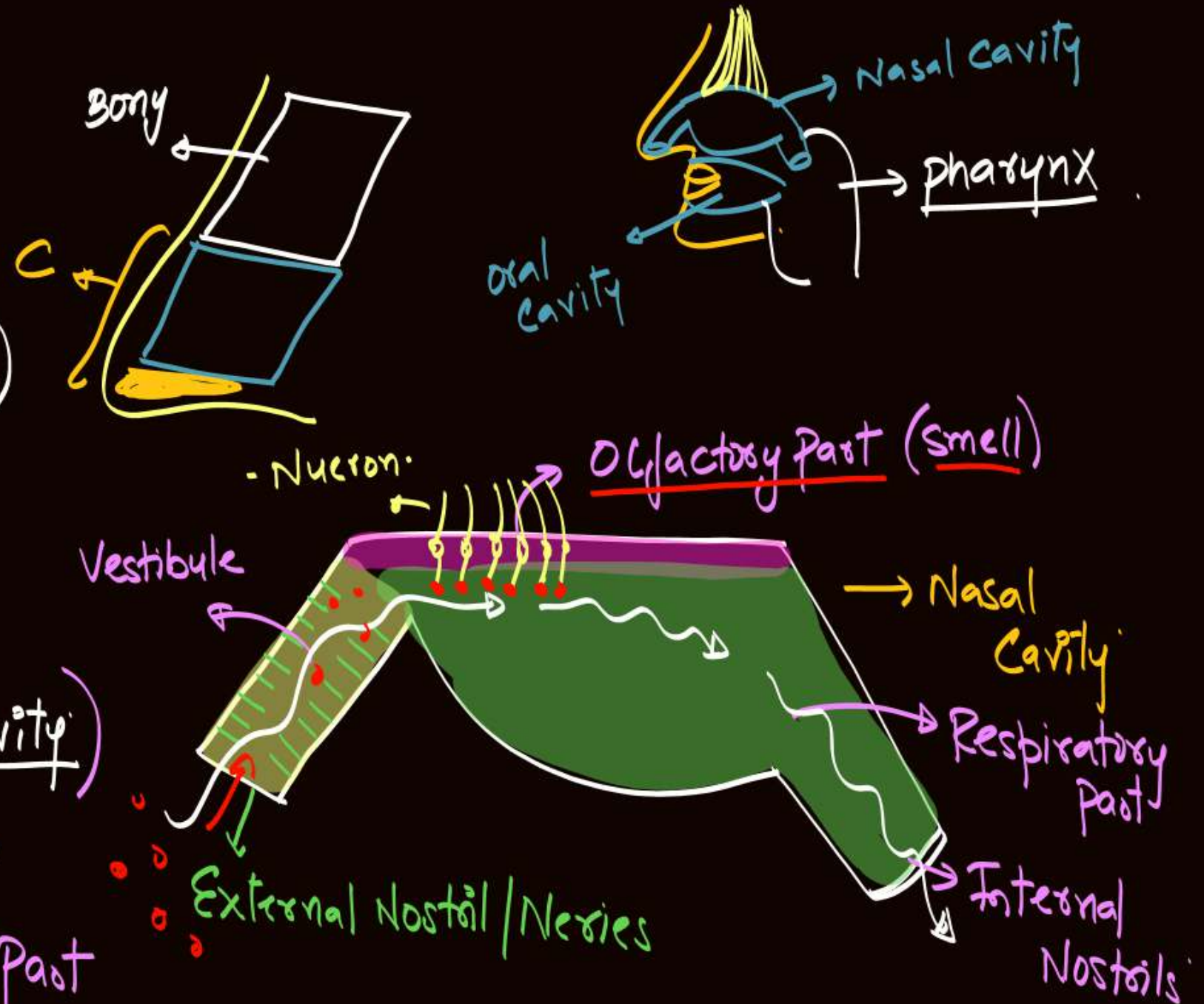
(b) Tip of Nose
(Elastic Cartilage)

→ Cavity inside Nose, is called (Nasal Cavity)

→ Nasal cavity has 3-parts (a) vestibule.

(b) Olfactory Part

(c) Respiratory Part



Function of Nose

- ① Passage to Air
- ② Purifies the Air
- ③ Traps Dirt with the help of Mucous
- ④ Helps in olfaction (Smell)
- ⑤ Warms the Air

(5 Sense organs)

QUESTIONS



+human.

Which is the correct sequence of air passage in man?

- 1 ✓ External nostril → Nasal passage → Internal nostril → Pharynx → Larynx → Trachea → Bronchi → Bronchioles → Alveoli
- 2 ✗ Nose → Larynx → Pharynx → Bronchioles → Bronchi → Alveoli
- 3 Nose → Pharynx → Larynx → Bronchioles → Bronchi → Alveoli
- 4 Nose → Bronchioles → Pharynx → Larynx → Bronchi → Alveoli

QUESTIONS



Pulmonary respiration is most well developed in

(Lungs).

(Great)

- 1 Mammals
- 2 Invertebrates
- 3 Amphibians
- 4 Hemichordates



QUESTIONS

Earthworm respire through

- 1 Moist skin
- 2 Ctenidia
- 3 Clitellum
- 4 Typhlosole

Thank
You